

Assignment III

① $\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$

$\sigma_y | \text{eigenvector} \rangle = \text{eigenvalue} | \text{eigenvector} \rangle$

$$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \cdot x + (-i) \cdot y \\ i \cdot x + 0 \cdot y \end{pmatrix} = \begin{pmatrix} -iy \\ ix \end{pmatrix}$$

$$\begin{pmatrix} -iy \\ ix \end{pmatrix} = \alpha \begin{pmatrix} x \\ y \end{pmatrix} \quad \xrightarrow{\text{eigenvalue}}$$

$$-iy = \alpha x \quad \text{and} \quad ix = \alpha y$$

$$-i \cancel{y} = \alpha \left(\frac{\alpha \cancel{y}}{i} \right)$$

$$-i \cdot i = \alpha^2$$

$$\alpha^2 = 1$$

$$\alpha = \pm 1$$

when $\alpha = 1$ we know 1st vector is x part of matrix $-i$ and y part of matrix is 1 .

$$= \frac{1}{\sqrt{2}} \begin{pmatrix} -i \\ 1 \end{pmatrix}$$

when $\alpha = -1$, $\frac{1}{\sqrt{2}} \begin{pmatrix} i \\ 1 \end{pmatrix}$

② $\langle 0 | \sigma_y$

$$\begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \begin{pmatrix} 1 \cdot 0 + 0 \cdot i & 1 \cdot (-i) + 0 \cdot 0 \end{pmatrix} = \begin{pmatrix} 0 & -i \end{pmatrix}$$

$$\sigma_y | 0 \rangle = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \cdot 1 + (-i) \cdot 0 \\ i \cdot 1 + 0 \cdot 0 \end{pmatrix} = \begin{pmatrix} 0 \\ i \end{pmatrix}$$

③ $\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$

$$\sigma_y^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 - i^2 & 0 + 0 \\ 0 + 0 & -i^2 + 0 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

$$\sigma_y^2 = I$$

④ $\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

$$\sigma_z^2 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} 1+0 & 0+0 \\ 0+0 & 0+1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

⑤ $\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$

$$\sigma_y^\dagger = \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}$$

$$\sigma_y^T = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

σ_y is Hermitian matrix.